MSDM 5058 Information Science
Assignment 2 (due 14th March, 2026)

Submit your assignment solution on Canvas. You may discuss with others or seek help from your TA, but you should not directly copy from others. Otherwise, it will be considered as plagiarism.

(1) Average entropy

Let $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ be the entropy function of a binary source.

(a) Evaluate $H(1/3)$.

(b) Calculate the expected entropy $H(p)$ when the probability p is chosen uniformly in the range $0 \le p \le 1$.

Solution:

(a)

$$H\left(\frac{1}{3}\right) = -\frac{1}{3}\log_2\frac{1}{3} - \left(1-\frac{1}{3}\right)\log_2\left(1-\frac{1}{3}\right) = \log_2 3 - \frac{2}{3} = 0.918 \text{ bits}$$

(b)

$$\begin{aligned} H(p) &= \frac{\int_0^1 [-p\log_2 p - (1-p)\log_2(1-p)]dp}{1-0} \\ &= -\int_0^1 \frac{p\ln p + (1-p)\ln(1-p)}{\ln 2} dp = -\frac{2}{\ln 2}\int_0^1 x\ln x\, dx \\ &= -\frac{2}{\ln 2}\left(\frac{x^2}{2}\ln x - \frac{x^2}{4}\right)\Bigg|_0^1 = \frac{1}{2\ln 2} = 0.721 \text{ bits} \end{aligned}$$

(2) Mutual information for correlated normal distributions

X and Y are two correlated random normal variables with the following joint probability distribution

$$\binom{X}{Y} \sim N_2\left(0, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix}\right)$$

Evaluate $I(X;Y)$ and comment on the cases when $\rho = 1,\ 0$, and -1.

Solution:

The marginal probability distribution of X is $N(0, \sigma^2)$, i.e.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}$$

Its entropy is

$$\begin{aligned}
H(X) &= -\int_{-\infty}^{\infty} f(x) \ln f(x)\ dx \\
&= -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \left[-\frac{x^2}{2\sigma^2} - \ln\sqrt{2\pi\sigma^2}\right] dx \\
&= \frac{1}{\sqrt{\pi(2\sigma^2)^3}} \underbrace{\int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2\sigma^2}} dx}_{\frac{1}{2}\sqrt{\pi(2\sigma^2)^3}} + \frac{\ln\sqrt{2\pi\sigma^2}}{\sqrt{2\pi\sigma^2}} \underbrace{\int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}} dx}_{\sqrt{2\pi\sigma^2}} \\
&= \frac{1}{2} + \frac{1}{2}\ln(2\pi\sigma^2)
\end{aligned}$$

The joint probability distribution function of X and Y is

$$f(x,y) = \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{-\frac{(x^2+y^2)-2\rho xy}{2(1-\rho^2)\sigma^2}}$$

Their joint entropy is

$$\begin{aligned}
H(X,Y) &= -\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y) \ln f(x,y)\, dx\, dy \\
&= -\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{-\frac{(x^2+y^2)-2\rho xy}{2(1-\rho^2)\sigma^2}} \left[-\frac{(x^2+y^2)-2\rho xy}{2(1-\rho^2)\sigma^2}\right. \\
&\qquad \left. - \ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right)\right] dx\, dy
\end{aligned}$$

We now separate it into two parts. The integral involving the first term in the square brackets is found by letting $u = \frac{x-\rho y}{\sqrt{1-\rho^2}}$ and $v = y$; in other words, $x = u\sqrt{1-\rho^2} + v\rho$.

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{-\frac{(x^2+y^2)-2\rho xy}{2(1-\rho^2)\sigma^2}} \left[\frac{(x^2+y^2)-2\rho xy}{2(1-\rho^2)\sigma^2}\right] dx\, dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{\frac{(x-\rho y)^2+y^2(1-\rho^2)}{2(1-\rho^2)\sigma^2}} \left[\frac{(x-\rho y)^2+y^2(1-\rho^2)}{2(1-\rho^2)\sigma^2}\right] dx\, dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{\frac{u^2+y^2}{2\sigma^2}} \left(\frac{u^2+v^2}{2\sigma^2}\right) \underbrace{\left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\right)}_{\sqrt{1-\rho^2}} du\, dv$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2} e^{\frac{u^2+y^2}{2\sigma^2}} \left(\frac{u^2+v^2}{2\sigma^2}\right) du\, dv$$

Then, let $u = r\cos\theta$ and $v = \sin\theta$ so that $u^2 + v^2 = r^2$.

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2} e^{\frac{u^2+v^2}{2\sigma^2}} \left(\frac{u^2+v^2}{2\sigma^2}\right) du\, dv$$

$$= \frac{1}{2\pi\sigma^2}\int_0^{2\pi}\int_0^{\infty} e^{\frac{r^2}{2\sigma^2}} \frac{r^2}{2\sigma^2} \underbrace{\left(\frac{\partial u}{\partial r}\frac{\partial v}{\partial \theta} - \frac{\partial u}{\partial \theta}\frac{\partial v}{\partial r}\right)}_{r} dr\, d\theta$$

$$= \frac{1}{2\sigma^4}\int_0^{\infty} r^3 e^{\frac{r^2}{2\sigma^2}} dr = \frac{1}{2\sigma^4}\frac{1!}{2\left(\frac{1}{2\sigma^2}\right)^2} = 1$$

The integral involving the second term in the square brackets similarly becomes

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{-\frac{(x^2+y^2)-2\rho xy}{2(1-\rho^2)\sigma^2}} \left[\ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right)\right] dx\, dy$$

$$= \left[\ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right)\right]\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{-\frac{(x-\rho y)^2+y^2(1-\rho^2)}{2(1-\rho^2)\sigma^2}} dx\, dy$$

$$= \left[\ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right)\right]\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2} e^{\frac{u^2+v^2}{2\sigma^2}} du\, dv$$

$$= \frac{\ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right)}{2\pi\sigma^2}\int_0^{2\pi}\int_0^{\infty} r e^{\frac{r^2}{2\sigma^2}} dr\, d\theta$$

$$= \frac{\ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right)}{\sigma^2}\frac{1!}{2\left(\frac{1}{2\sigma^2}\right)} = \ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right)$$

Therefore, $H(X,Y) = 1 + \ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right)$ and

$$\begin{aligned} I(X;Y) &= H(X) + H(Y) - H(X,Y) \\ &= \frac{1}{2} + \frac{1}{2}\ln(2\pi\sigma^2) + \frac{1}{2} + \frac{1}{2}\ln(2\pi\sigma^2) - 1 - \ln\left(2\pi\sigma^2\sqrt{1-\rho^2}\right) \\ &= -\frac{1}{2}\ln(1-\rho^2) \end{aligned}$$

- $\rho = 1$: The mutual information becomes infinite because means knowing X implies perfect knowledge about Y.
- $\rho = 0$: The mutual information becomes zero because in this case, X and Y are independent of each other.
- $\rho = -1$: The mutual information becomes infinite because, again, knowing X implies perfect knowledge about Y.

(3) Channel capacity

Consider a binary asymmetric communication channel. Its input alphabet is $X = \{0, 1\}$ with probabilities $\{0.5, 0.5\}$, and its output alphabet is $Y = \{0, 1\}$. Its channel matrix is

$$\begin{pmatrix} 1-\alpha & \beta \\ \alpha & 1-\beta \end{pmatrix}$$

where α is the probability of transmission error when sending $X = 0$ and β is the probability of transmission error when sending $X = 1$.

(a) What is the entropy $H(X)$ of the input?

(b) What is the probability distribution $p(Y)$ of the outputs and the entropy $H(Y)$ of this output distribution?

(c) What is the joint probability distribution $p(X, Y)$ of the input and the output? What is their joint entropy $H(X, Y)$?

(d) What is the mutual information $I(X; Y)$ of this channel as a function of α and β?

(e) How many combinations of (α, β) are there for which the mutual information of this channel is maximal? What are those values, and what then is the capacity of such a channel in bits?

(f) What condition does (α, β) satisfy when the capacity of this channel is minimal? What is the channel capacity in that case?

Solution:

(a) The entropy of the input is $H(X) = -0.5 \log_2 0.5 - (1 - 0.5) \log_2(1 - 0.5) = 1$ bit.

(b) The probability distribution of the outputs are

$$p(y = 0) = 0.5(1 - \alpha + \beta)$$
$$p(y = 1) = 0.5(\alpha + 1 - \beta)$$

The entropy of this distribution is therefore

$$\begin{aligned} H(Y) &= 1 - 0.5(1 - \alpha + \beta) \log_2(1 - \alpha + \beta) - 0.5(\alpha + 1 - \beta)\log_2(\alpha + 1 - \beta) \\ &= 1 - 0.5\alpha \log_2(\alpha + 1 - \beta) - 0.5(1 - \alpha) \log_2(1 - \alpha + \beta) \\ &\quad - 0.5\beta \log_2(1 - \alpha + \beta) - 0.5(1 - \beta) \log_2(\alpha + 1 - \beta) \end{aligned}$$

(c) The joint probability distribution is

$$p(X,Y) = \begin{pmatrix} 0.5(1-\alpha) & 0.5\beta \\ 0.5\alpha & 0.5(1-\beta) \end{pmatrix}$$

The entropy of this joint distribution is

$$\begin{aligned} H(X,Y) &= -\sum_{x,y} p(x,y)\log_2 p(x,y) \\ &= 1 - 0.5\alpha\log_2\alpha - 0.5(1-\alpha)\log_2(1-\alpha) - 0.5\beta\log_2\beta \\ &\quad - 0.5(1-\beta)\log_2(1-\beta) \end{aligned}$$

(d) The mutual information is $I(X;Y) = H(X) + H(Y) - H(X,Y)$. From the above, we have

$$\begin{aligned} I(X;Y) = 1 &- 0.5\alpha\log_2\left(\frac{\alpha+1-\beta}{\alpha}\right) - 0.5(1-\alpha)\log_2\left(\frac{1-\alpha+\beta}{1-\alpha}\right) \\ &- 0.5\beta\log_2\left(\frac{1-\alpha+\beta}{\beta}\right) - 0.5(1-\beta)\log_2\left(\frac{\alpha+1-\beta}{1-\beta}\right) \end{aligned}$$

(e) When $(\alpha,\beta) = (0,0)$ or $(1,1)$ (corresponding to perfect transmission and perfectly erroneous transmission respectively), the mutual information reaches its maximum of 1 bit, which is also the channel capacity.

(f) When $\alpha + \beta = 1$, the channel capacity reaches its minimum and is equal to 0.

(4) Shannon, Fano, and Huffman codes

A source X has an alphabet of seven characters $\{a,b,c,d,e,f,g\}$ with probabilities $\{0.01, 0.24, 0.05, 0.20, 0.47, 0.01, 0.02\}$.

(a) What is the entropy of X?

(b) Give an example of a set of codewords for X using Shannon code, Fano code, and Huffman code respectively and explain how you get it.

(c) What are the average lengths of the three sets of codewords obtained in (b)?

(d) Which code is the optimal code for X and how much greater is its expected length than the entropy of X?

Solution:

(a) $H(X) = -\sum_i p_i \log_2 p_i = 1.932$ bits.

(b) An example of the Shannon code is

$$\{a{:}\,1000110;\ b{:}\,010;\ c{:}\,10000;\ d{:}\,011;\ e{:}\,00;\ f{:}\,1000111;\ g{:}\,100010\}$$

It is obtained by giving each codeword a length of $\left\lceil \log_2 \frac{1}{p_i} \right\rceil$.

An example of the Fano code is

$$\{a\!: 111110;\ b\!: 10;\ c\!: 1110;\ d\!: 110;\ e\!: 0;\ f\!: 111111;\ g\!: 11110\}$$

It is obtained by building a tree and dividing each of its branches iteratively into two subbranches with approximately equal chance. For example, here we first put e in the first branch and the other six characters in the second branch. We then continue the division until each character is assigned a codeword.

An example of the Huffman code is

$$\{a\!: 000000;\ b\!: 01;\ c\!: 0001;\ d\!: 001;\ e\!: 1;\ f\!: 000001;\ g\!: 00001\}$$

It is obtained by iteratively grouping the two items with the smallest probability. For example, here we first group a and f together and assign them with the longest codewords, which are 000000 and 000001. We combine them as a new item with a probability of 0.02 and then continue the procedure.

(c) For the Shannon code, the lengths of the codewords are 7, 3, 5, 3, 2, 7, and 6, so the expected length is 2.77 bits.
For the Fano code, the lengths of the codewords are 6, 2, 4, 3, 1, 6, and 5, so the expected length is 1.97 bits.
For the Huffman code, the lengths of the codewords are also 6, 2, 4, 3, 1, 6, and 5, so the expected length is 1.97 bits.

(d) Both Fano and Huffman codes are optimal for X. Their expected length is greater than the entropy of X by $1.970 - 1.932 = 0.038$ bits.